CCCA Cell-State Analysis — Rendeiro et al. 2020

Chronic Lymphocytic Leukemia · 10x · 18,314 malignant B-cells · 7 samples / 4 patients

GO / NO-GO GATES

Gate 1 — mixing & continuity

GO

Harmony (θ=10) fuses the four patient-private CLL clones: per-neighbourhood iLISI 1.0→~5.1 / 7 across a θ×n_pcs sweep (plateau at θ=10; θ=20 over-collapses), one connected kNN component after island removal. Judged on the regenerated embedding — patients intermix on one continuous body. Decisive check: the Gast→Crest gradient reproduces WITHIN each clone (Crest↔Gast −0.47…−0.58 in every patient), so integration aligns a shared intra-clonal gradient, not a fabricated continuum; proliferation diffuse (not a cell-cycle confound, corr≈0).

Gate 2 — module geography

GO (non-branching)

Signatures collapse onto a single (Gast/pSom)→Crest axis: raw per-cell Crest↔Gast r=−0.55, Crest↔pSom r=−0.55, Gast↔pSom r≈+0.05 (no cross-repression → no two competing arms). Crest is an inverse terminal gradient. Principal curve is fold-free (2 tips / 0 forks; self-approach 1.6). Crest is NOT enriched on a Gast arm — no inversion flag.

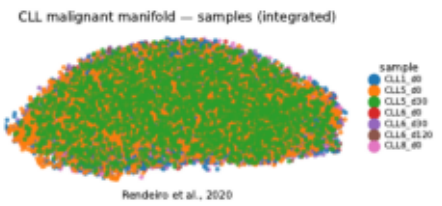
Gate 3 — hub cross-repression

NA

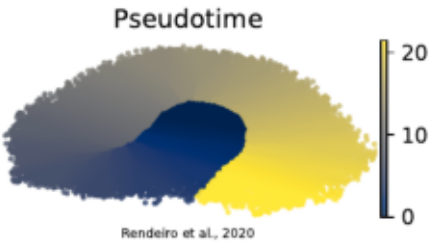
Non-branching topology — no bifurcation, no Gast/pSom hub, so the branching-only cross-repression test does not apply. PPT tree gives only unstable spurious forks (4→6→9 as nodes 100→200) over a blob. Modules notebooks not produced.

STEP FIGURES

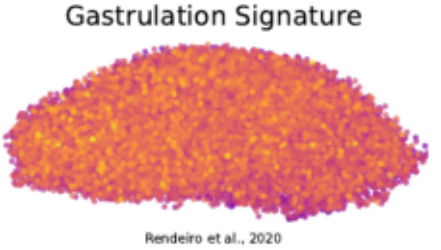
01 · Samples (integration)



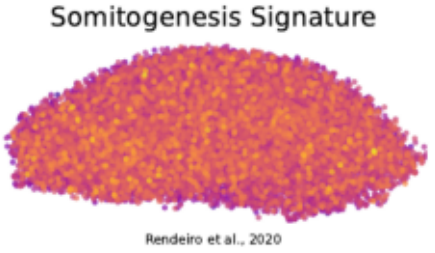
02 · Pseudotime (curve)



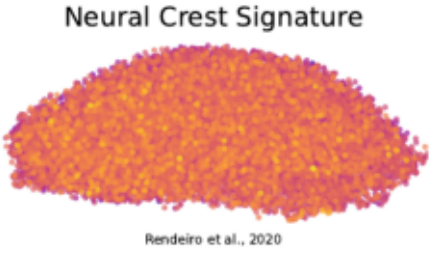
03 · Gastrulation



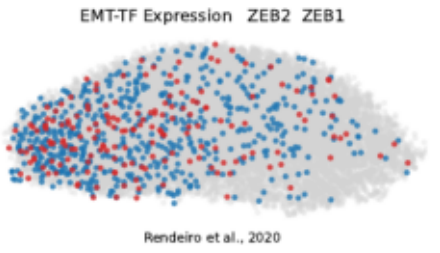
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs (ZEB2, ZEB1)



Dataset: Chronic lymphocytic leukemia (Rendeiro 2020), 10x, 18,314 malignant B-cells over 7 samples / 4 patients.

Verdict: Pattern reproduces as the NON-BRANCHING case — a single gastrulation/somitogenesis→neural-crest developmental axis; no hub, no bifurcation.

PARAMETER CHOICES & RATIONALE

Min-cell threshold	≥285 — geometric mean of the 1012→80 spillover cliff (ratio 12.6×); keeps 7/11 samples, 4 patients
Integration	Harmony on sample, θ=10 (strong; patient-private CLL clones), n_pcs=30, spread=1.3
Islands removed	343 cells / 57 detached specks (UMAP 6× radius graph, largest component)
Geometry	Non-branching → scFates principal curve (ElPiGraph), Nodes=20; 2 tips / 0 forks, fold-free (self-approach 1.6)
Root node	Curve tip 2 — Gast-high, Crest-low (developmental origin; Crest terminal along pseudotime)
Module means (raw)	Gast −0.017 · pSom +0.007 · Crest +0.007 → Linear
EMT-TFs shown	ZEB2 (12.1%) + ZEB1 (3.8%) — only factors clearing the 3% detection floor (Snail/Twist/FOXC/PRRX silent)